WAP to initialize the non static fields using parameter variable as per requirements :
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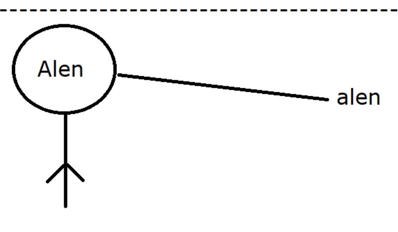


```java
package com.ravi.oop_ex;

//ELC
public class EmployeeDemo
{
    public static void main(String[] args)
    {
        Employee alen = new Employee();
        int id = Integer.parseInt(IO.readln("Enter employee id :"));
        String name = IO.readln("Enter employee Name :");
        double salary = Double.parseDouble(IO.readln("Enter employee Salary :"));

        alen.setEmployeeData(id, name,salary);
        alen.setEmployeeGrade();
        String result = alen.getEmployeeData();
        IO.println(result);
    }
}

//BLC
class Employee
{
    int employeeId;
    String employeeName;
    double employeeSalary;
    char employeeGrade;

    public void setEmployeeData(int id, String name, double salary)
    {
        employeeId = id;
        employeeName = name;
        employeeSalary = salary;
    }

    //Initialize employee grade based on the employee salary
    public void setEmployeeGrade()
    {
        if(employeeSalary>=100000)
        {
            employeeGrade = 'A';
        }
        else if(employeeSalary >=75000)
        {
            employeeGrade = 'B';
        }
        else if(employeeSalary >=50000)
        {
            employeeGrade = 'C';
        }
        else
        {
            employeeGrade = 'D';
        }
    }



    public String getEmployeeData()
    {
        return "[Employee id is :"+employeeId+", Name is :"+employeeName+", Salary is 
:"+employeeSalary+", Grade is :"+employeeGrade+"]";
    }
}
```

WAP to show the Manager details, the bonus amount of manager depends upon the number of projects based on the following criteria :

```java
package com.ravi.oop_ex;

public class ManagerDemo
{
   void main()
   {
    Manager smith = new Manager();
    smith.setManagerData(102, "Mr Smith", "Hyderabad", 120000);
    smith.getManagerData();
    int noOfProj = Integer.parseInt(IO.readln("Enter no of projects :"));
    smith.calculateBonusAmount(noOfProj);

   }
}

class Manager
{
    int managerId;
    String managerName;
    String managerAddress;
    double managerSalary;


    public void setManagerData(int id, String name, String address, double salary)
    {
        managerId = id;
        managerName = name;
        managerAddress = address;
        managerSalary = salary;
    }

    public void getManagerData()
    {
        IO.println("Manager Id is :"+managerId);
        IO.println("Manager Name is :"+managerName);
        IO.println("Manager Address is :"+managerAddress);
        IO.println("Manager Salary is :"+managerSalary);
    }

    public void calculateBonusAmount(int noOfProjects)
    {
        double bonus = 0.0;

        if(noOfProjects >=1 && noOfProjects <=5)
        {
            bonus = managerSalary * 0.20;
        }
        else if(noOfProjects >=6 && noOfProjects <=10)
        {
            bonus = managerSalary * 0.40;
        }
        else
        {
            bonus = 0.0;
        }

        IO.println("Your no of projects are :"+noOfProjects);
        IO.println("Your bonus amount is :"+bonus);
        IO.println("This month final salary is :"+(managerSalary+bonus));

    }

}
```

Note : Up to here we have learned two ways to initialize the non static fields

1) By using reference variable
   Example : raj.rollNumber = 101;

2) By using parameter variable [Park story]
   Example:
   raj.setStudentData(101, "Raj", 85);

   These values are recieved by parameter variable to initialize the non static fields

**What is a constructor [Introduction only]**
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* If the **name of the class** and **name of the method** both are **exactly same** and it does not contain **any return type then It is called constructor.**
  Example :
  public class **Employee**
  {
    public **Employee**() //Constructor
    {
    }

    public void Employee()  //Method
    {
    }
  }

* In Java, Whenever we write a class (either BLC OR ELC) and as a programmer, if we don't write any type of constructor then automatically **java compiler will add one default no argument constructor in that particular class.**
  Example :

```
public class Employee
{
   int id;
   String name;
   double salary;

}
```

javac

```
public class Employee
{
  int id;
  String name;
  double salary;

   public Employee() //default no argument constructor [added by]
   {                                                      javac
   }
}
```

* The access modifier of default no argument constractor (added by java compiler) would be same as class access modifier means, if class is public, access modifier of default constructor is also public, if class is non public then access modifier is also non public.

* Command to show that default no argument constrcutor is added by java compiler :